15.1 Stages of the project life cycle

A well-managed project ensures deadlines are met, **resources** are available, and that each person knows the job that they are meant to do.

The project management process is usually carried out in four main stages: project initiation, planning, execution and monitoring, and project close, as shown in Figure 15.2.

Figure 15.2: Project life cycle stages.

Project initiation

The project **initiation** stage sets the goals of the project. This defines what the project will do, and why there is a need for the project. A number of aspects need to be determined during the project initiation stage.

Objectives

The project initiation stage identifies objectives. Each project will have set objectives. For example, a project may be to install a new IT system for a business. Each large objective may be split into smaller objectives. Installing an IT system could be divided into:

1. purchase the equipment
2. prepare the building (cabling and power sources)
3. install the new machines

4. check that the machines work
5. ensure users are satisfied.

Each of these objectives could then be subdivided into even smaller objectives. It is important to list all the objectives at the start of the project.

All projects will have a person who is called the owner of the project. This is usually the person who wants the project to be completed. The owner could be an external customer, or an internal member of staff. The owner of the project will usually agree to the objectives at the start. This means that they agree that all the objectives they want to achieve have been listed.

Resources

The initiation stage identifies the resources required for the project. Resources may be items like new computers or cables. Resources can also be people such as IT technicians or electricians. Time is also a resource. Resources usually cost money. Projects can run over time and money if all resources are not identified at the start. This is because more resources need to be bought later.

Success criteria

Success criteria are essential elements of the project that must be completed. These are not the same as goals. For example, the success criteria for installing an IT system could include 'Must have enough machines for 150 people', 'Must use Windows OS' or 'Must have 2 monitors per machine'.

Stakeholders and their needs

Stakeholders are anyone who has an interest in the project (see Figure 15.3). For example, a stakeholder could be the user of the new IT system. Stakeholders are important as they comment on the success criteria and say what they want from the project. They can impact the goals and the success criteria.

Project scope and high-level schedules

Project scope defines how 'large' the project is. In big projects, some tasks may need to be outsourced to other businesses. For example, connecting a new electricity supply to a building for the IT system needs to be completed by the local electricity company. Connecting the electricity is called 'out of scope' as this is not something to include in the project.

Figure 15.3: Stakeholders may include the end-users who will be using the new IT system.

A high-level project schedule sets out a rough plan of the tasks involved and how long they will take. This will be based on the resources, goals and success criteria.

Project planning

DID YOU KNOW?

There is a common phrase: 'failing to prepare is preparing to fail'. This means that if you do not plan something fully, then it is most likely to fail.

Projects are planned so that all stakeholders know what to do during all stages of the project. Budgets and schedules are created for the project. Milestones will be set. Milestones are dates by each part of the project must be completed. Tasks for one milestone are done first, before the tasks of milestones that come later. A **Gantt chart** is produced to show the order in which tasks should be completed.

Resources are allocated to tasks. Resources need to be available at the time they are needed. For example, an electrician will be allocated to the project at the time when the electrical work needs to be completed. The most important part of planning ensures that appropriate time is allocated to each task, and resources for each task are organised so that tasks can be completed on time.

It is also wasteful to have resources not being used during the project. For example, you would not hire an electrician to install the software for the IT system. You may also group the tasks for the electrician into one milestone. This means that you can hire the electrician once, and complete all the tasks. For example, they could connect the new power supply, and fit all of the new wiring that may be needed. Hiring an electrician many times for short periods of time is inefficient and may cost more money.

Project execution and monitoring

Implementing the plan

Once the project plans are completed, a start date to begin to implement the plans is set. The project starts on the start date. The **execution** of a project is the 'doing' part. This is when the milestones are completed.

Implementation of the project plan completes the stages and goals listed in the project plan.

Monitoring progress against time

As each task is completed it is monitored. Monitoring checks that each part of the project is going to be completed on schedule (on time). It is important that the plan is followed closely. This helps to make sure that resources are used at the appropriate times. Delays to tasks will mean that tasks that are scheduled to occur later will also be delayed.

Each project will have a project manager. The project manager informs each member of the project team about their role, their responsibilities and the milestone dates by which they are expected to have completed their tasks. The project manager is responsible for keeping the project to schedule (see Figure 15.4).

Projects are not always completed on time. This is often because smaller tasks may take longer to complete and milestones may not be met. Each delay affects the next section of the project, and so the delays increase as the project moves forward. However, good project monitoring can identify where delays may occur. This allows a project manager to re-plan small sections of the project to help ensure the whole project remains on schedule.

The project manager also monitors the performance of the project team to ensure they carry out their roles effectively. The project manager monitors expenditure and compares it against the budget. This helps to ensure that the project does not spend more than its allocated budget. The project manager also ensures the scope of the project does not grow beyond the agreed boundaries.

Cost and quality

A project manager will check that the costs of the project stay within the budget that was set. Sometimes this may not be possible. For example, during the COVID-19 pandemic, there was a shortage of processors and as a result the cost of processors rose quickly, so installing an IT system became more expensive. It is important that increases in costs are discussed with the stakeholders. If the project becomes too expensive, it may need to be cancelled.

Alternatively, costs may go down. For example, a different electrician may be able to complete the jobs required for less money.

A project will also be monitored for quality. Each task will be checked to ensure it is completed well. For example, the work of the electrician will be checked to ensure that there are enough cables and the cabling is installed correctly. It is important to monitor the quality of the tasks completed because it will often cost more money and delay the project if mistakes are only noticed later in the project life cycle.

Figure 15.4: All aspects of the project will be monitored to ensure work is completed on time.

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Reporting to stakeholders

The stakeholders in a project will want to know how the project is progressing in terms of:

- time
- cost
- delays
- quality.

During the project execution stage, the project manager will prepare reports to give to the stakeholders. (see Figure 15.5) This allows them to monitor the progress of the project and provide input if needed.

Figure 15.5: A project manager will monitor the project as it progresses.

Project close

A handover happens between the project team and the client when the project is complete. The client reviews the project with the project manager. If the client is happy that everything is finished as expected then the client signs off the project as completed. When the client signs the project off it means that there is no further work to carry out. When it is signed off then any contracts will be closed with the workers. Any resources will also be released. This means that they are no longer needed and other people or projects can use them.

At the end of the project monitoring and execution stage, a review takes place. The review will look at what went wrong and what went well. The review will cover questions such as:

- Was the project schedule maintained?
- Was the project completed within budget?
- Were all resources available when required?
- Has the project met the original requirements?

The review of the project will be stored. Reviews are used to help plan future projects. The reviews will help projects avoid the same mistakes that may have happened in other projects.

Questions

1 Identify three things that project initiation will find.
2 State the connection between resources and tasks.
3 What does a Gantt chart show?
4 Explain why monitoring progress is needed in a project.
5 Explain why the project manager may need to re-plan parts of the project.
6 State the risk of not monitoring the quality of completed tasks.
7 Give two examples of review questions.